

PES UNIVERSITY, BANGALORE

Department of Computer Science and Engineering

Smart Complaint Management System

Campus Facility Complaint Registration, Routing and Resolution Tracking

SOFTWARE REQUIREMENTS SPECIFICATION

(Deliverable 2)

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Team Profile

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1. Introduction

1.1 Purpose

This document specifies the software requirements for the **Smart Complaint Management System (SCMS), Release 1.0** — a role-based web application for registering, routing, resolving and tracking campus facility and hostel complaints at PES University.

This SRS covers the **complete scope of Release 1.0** and is not limited to a subsystem. It specifies all functional behaviour, external interfaces, non-functional requirements and business rules that Release 1.0 must satisfy. Functionality explicitly excluded from Release 1.0 is listed in Section 1.3.

1.2 Intended Audience and Reading Suggestions

This document is written for the project team, the course faculty who evaluate the deliverable, and the Campus Facility and Hostel Administration Office as customer. Developers should read Section 5 and the appendices for implementable detail; evaluators and the customer should read Section 1 to Section 4 for scope and context, then Section 6 for the quality obligations.

Organisation of the rest of this document. Section 2 gives the product functions, user classes, operating environment, constraints and assumptions. Section 3 specifies the external interfaces. Section 4 contains the analysis models. Section 5 contains the functional requirements organised by system feature, each with a uniquely tagged requirement list. Section 6 contains the performance, safety, security, quality and business-rule requirements. The appendices provide the glossary, the field-level data requirements and the requirement traceability matrix.

1.3 Product Scope

SCMS replaces the current informal, unrecorded campus complaint process with a recorded, automatically routed and time-bound workflow that is visible end-to-end to complainant, resolver and administrator.

Objectives of Release 1.0

1. Every complaint is recorded and carries a unique, quotable identifier from the moment of submission.
2. Every complaint is routed to a responsible department automatically from administrator-configured rules, with no manual forwarding step.
3. Every complaint carries a resolution time target, and any complaint breaching that target is automatically escalated and made visible.
4. Every state change is recorded with actor, timestamp and remarks, producing a complete audit trail.
5. The complainant can track progress at any time, is notified of state changes, and can confirm or reject the resolution before closure; if no response is received within the configured confirmation period, the system closes the complaint automatically and records the closure as unconfirmed.
6. Administrators obtain consolidated reporting on volume, resolution performance and overdue work.

Explicitly outside the scope of Release 1.0

Machine-learning or NLP complaint classification, sentiment analysis and chatbots; AI duplicate detection and predictive maintenance analytics; native mobile applications; SMS/WhatsApp gateways; live chat, voice or video; IoT fault-detection hardware; payment, billing or materials procurement; integration with university ERP/HRMS/academic systems; third-party single sign-on and biometric authentication; multi-campus tenancy; geospatial maps and GPS location capture; staff attendance or shift rostering; offline operation; multilingual interfaces.

2. Overall Description

2.1 Product Functions

At a high level, SCMS must let its users:

- Register an account and authenticate; act only within the permissions of an assigned role.
- Register a complaint with category, location, description and priority indication, optionally attaching photographic or document evidence, and receive a unique complaint identifier.
- Submit a complaint anonymously and track it using a system-issued tracking reference.
- Have the complaint routed automatically to a responsible department from configured category-and-location rules, and placed in that department's work queue.

- Have a resolution time target derived from the complaint's priority, and have the complaint automatically escalated when that target is breached.
- Work a complaint through a controlled status lifecycle in which only valid, role-authorised transitions are permitted, with every transition recorded in an immutable audit trail.
- Record resolution remarks and resolution evidence against a complaint.
- Track a complaint's current status and full history, and be notified in-app and by email of significant state changes.
- Confirm a resolution and close the complaint, or reject it and cause the complaint to be reopened.
- Reassign a complaint to another staff member within the same department.
- Administer master data — categories, departments, staff membership, routing rules and SLA thresholds.
- Obtain consolidated reports on complaint volume, resolution performance and open, overdue and escalated complaints.

2.2 User Classes and Characteristics

User class	Frequency of use	Functions used	Technical expertise	Privilege level	Importance
Complainant — student, hostel resident, faculty, non-teaching staff	Occasional but time-sensitive	Register, log in, submit complaint, attach evidence, track own complaints, receive notifications, confirm/reject resolution, reopen	Low; non-technical; predominantly mobile browser	Lowest	**Most important** — largest population; the system fails if this class cannot use it unaided
Anonymous Complainant	Rare	Submit complaint without login; track by tracking reference only	Low	Lowest; unauthenticated	Important — exists specifically to remove the attribution fear that suppresses reporting
Department Staff / Resolver — electrician, plumber, IT technician, housekeeping supervisor	Frequent, short interactions, multiple times daily	View own department queue, accept complaint, update status, record resolution remarks and evidence	Low to moderate; values speed and clarity over richness	Elevated, scoped to own department	**Most important** — all resolution throughput passes through this class
Department Head / Warden	Daily to weekly	Monitor department open / overdue / escalated complaints, reassign within department, receive SLA-breach notifications	Moderate	Elevated, supervisory, scoped to own department	Important — accountable for departmental performance
Administrator — Facility Administration Office	Weekly, plus initial setup	Manage users and roles, categories, departments, staff membership, routing rules, SLA thresholds; campus-wide reports	Moderate to high; trained	Highest	Important, but small in number

2.3 Operating Environment

Aspect	Specification
OE-1 Client platform	Any device with a modern browser — Android, iOS, Windows, macOS, Linux
OE-2 Client browsers	Current and immediately preceding major versions of Chrome, Firefox, Edge and Safari
OE-3 Client viewport	Must operate from 320 px width (small mobile) upward
OE-4 Server operating system	Linux (Ubuntu LTS or equivalent)
OE-5 Application runtime	Python 3.x with the Django framework
OE-6 Database	PostgreSQL in the deployed environment; SQLite permitted for local development only
OE-7 Containerisation	The application must run inside a Docker container
OE-8 Scheduler	An OS-level or container-level scheduler must be able to invoke the SLA escalation run periodically
OE-9 Mail	Reachable SMTP relay for outbound email
OE-10 Network	Campus network or internet access over HTTPS; the system must coexist with other applications on shared infrastructure and must not require exclusive use of the host

2.4 Design and Implementation Constraints

ID	Constraint
C-1	The implementation must use Python with Django.
C-2	Agile methodology only; requirements must be traceable to user stories on the team's Agile board.
C-3	Git must be used for version control with GitHub hosting; every team member must have traceable commits against their own functional feature.
C-4	A Continuous Integration (CI) pipeline (Jenkins) performing automated build, test and static analysis, code reviews and security validation are mandatory. No automated deployment is planned for Release 1.0.

ID	Constraint
C-5	Every requirement in this SRS must be traceable through architecture, design, code and test artefacts (Appendix C).
C-6	Static code analysis (SonarQube or equivalent) must be applied to the codebase.
C-7	Unit tests must report both line and branch coverage.
C-8	The system must be developed as four modular functional areas (F1–F4) with narrow interfaces, one owned end-to-end per team member.
C-9	No paid third-party service may be required for the system to function.
C-10	Any team member must be able to explain any submitted code, including LLM-assisted code.
C-11	Business logic must reside outside presentation templates so that it is separately unit-testable.
C-12	Adding a complaint category, department or routing rule must require configuration only, never a code change.

2.5 Assumptions and Dependencies

Assumptions (assumed factors, not established facts — if any is wrong, the affected requirements change)

ID	Assumption
D-A1	The Campus Facility & Hostel Administration Office acts as the customer; no external paying customer exists.
D-A2	The manual process characterised in the synopsis reflects actual current campus practice.
D-A3	Warden is a domain designation used in the hostel context for a person holding the Department Head role; it is not a separate fifth system role. The system defines exactly four roles: Complainant, Department Staff, Department Head and Administrator.
D-A4	Complaint categories, department mapping and SLA target durations used in this document are illustrative and require stakeholder confirmation.
D-A5	Users have working email addresses and can receive email from the deployment.
D-A6	Campus network availability is adequate; offline operation is not required.
D-A7	Deployment is single-campus and single-language (English).
D-A8	Expected load for Release 1.0 is of the order of a few thousand registered users and a few hundred complaints per month; concurrency is low.
D-A9	Physical resolution of a complaint happens outside the system; SCMS records the workflow, it does not perform the repair.

Dependencies

ID	Dependency
D-D1	Django framework and its authentication/permission subsystem
D-D2	Relational database engine (PostgreSQL)
D-D3	SMTP relay
D-D4	External scheduler
D-D5	Stakeholder availability for the confirmation interview
D-D6	Continuous Integration, coverage and static-analysis tooling

3. External Interface Requirements

3.1 User Interfaces

ID	Requirement
UI-1	All user interaction is through a responsive web interface rendered in a browser; the same interface must serve desktop and mobile viewports from 320 px width upward.
UI-2	Every authenticated screen must display a persistent header showing the logged-in user's identity, their role, a link to notifications with an unread count, and a logout control.
UI-3	Screens must be provided for: registration; login; complaint submission; complaint acknowledgement; "my complaints" list; complaint detail with status history; anonymous tracking lookup; staff work queue; complaint action/update; department monitoring view; master-data administration; reports dashboard.
UI-4	A first-time complainant must be able to complete registration and a full complaint submission traversing **no more than four primary screens**.
UI-5	Validation errors must be displayed adjacent to the offending field, must state what is wrong and what is required, and must preserve all other values the user has already entered.
UI-6	Error messages must never expose stack traces, internal identifiers, file paths, SQL fragments or framework debug output.
UI-7	Every complaint-detail screen must display the complaint identifier prominently and in a form that can be selected and copied.
UI-8	Status must be conveyed by text label in every case; colour may reinforce status but must never be the sole carrier of meaning.
UI-9	Any list that can exceed 25 rows must be paginated, and must offer filtering by status and by category.

ID	Requirement
UI-10	Destructive or irreversible actions (closing, rejecting, deleting master data) must require an explicit confirmation step.
UI-11	Detailed visual design is out of scope for this SRS and will be specified in the Phase-3 architecture/UI documentation.

3.2 Software Interfaces

ID	Interface	Direction and data	Purpose
SI-1	**Relational database** — PostgreSQL (deployed), SQLite (development only), accessed exclusively through the Django ORM	Bidirectional: users, roles, complaints, attachment metadata, categories, departments, routing rules, SLA configuration, status history, notifications	Persistent storage of all system data. Direct SQL from application code is prohibited; all access is through the ORM so that schema changes are migration-controlled.
SI-2	**SMTP relay** — accessed over SMTP/TLS using credentials supplied by configuration	Outbound only: recipient address, subject, body containing complaint identifier and current status	Delivery of the email notifications required by Section 5.8. No inbound mail processing exists in Release 1.0.
SI-3	**File storage** — server filesystem or an object store, addressed through the framework's storage abstraction	Outbound and inbound: uploaded evidence files	Storage of complaint attachments. File content is stored outside the database; only metadata and a storage reference are persisted.
SI-4	**Scheduler** — OS-level or container-level scheduler invoking a management command	Outbound trigger, no data payload	Periodic invocation of the SLA evaluation and escalation run (Section 5.6).
SI-5	**Static-analysis, coverage and Continuous Integration tooling**	Reads source and test artefacts	Satisfies constraints C-4, C-6 and C-7. Not part of the runtime system.
ID	Requirement on software interfaces		
SI-R1	All interface credentials and endpoints (database, SMTP, storage) must be supplied by environment configuration and must never be committed to version control.		
SI-R2	Failure of the SMTP interface must not cause loss of an in-app notification or failure of the underlying state transition (see REQ-NOTF-05).		
SI-R3	Failure of the file-storage interface during upload must cause the whole submission to fail atomically, leaving no partially created complaint (see REQ-CMP-07).		

3.3 Communications Interfaces

ID	Requirement
CI-1	All client–server communication must use HTTP over TLS (HTTPS); plain HTTP requests must be redirected to HTTPS.
CI-2	The application must serve standard HTML over HTTPS; forms must be submitted using HTTP POST for all state-changing operations. `GET` must never change server state.
CI-3	Every state-changing form submission must carry and validate an anti-CSRF token.
CI-4	Session identifiers must be transmitted in cookies marked `Secure`, `HttpOnly` and `SameSite`.
CI-5	Email must be transmitted to the SMTP relay over an encrypted channel (STARTTLS or SMTPS).
CI-6	Email notification bodies must contain the complaint identifier and current status but must **not** contain credentials, session tokens, or the anonymous tracking reference.
CI-7	File uploads must be transmitted as `multipart/form-data` over HTTPS and must be rejected at the boundary if they exceed the configured size limit.
CI-8	No inbound communication interface from any external system is exposed in Release 1.0.

4. Analysis Models

4.1 Actors

Actor	Type	Description
Complainant	Primary, human	Authenticated user who raises and tracks complaints

Actor	Type	Description
Anonymous Complainant	Primary, human	Unauthenticated user who raises a complaint and tracks it by reference
Department Staff	Primary, human	Resolves complaints assigned to their department
Department Head	Primary, human	Supervises a department's queue and reassigns work
Administrator	Primary, human	Configures master data and reads campus-wide reports
Scheduler	Secondary, system	Triggers the periodic runs that evaluate SLA breaches for escalation and close resolved complaints whose confirmation window has expired
Mail Service	Secondary, system	Delivers outbound email notifications

4.2 Use case inventory

UC ID	Use case	Primary actor	Realised by feature
UC-01	Register account	Complainant	5.1
UC-02	Log in / log out	All authenticated actors	5.1
UC-03	Submit complaint	Complainant	5.3
UC-04	Attach evidence to complaint	Complainant	5.3
UC-05	Submit anonymous complaint	Anonymous Complainant	5.4
UC-06	Track complaint by reference	Anonymous Complainant	5.4
UC-07	Route complaint to department	System, as part of the complaint submission flow	5.5
UC-08	Assign complaint to resolver	Department Head	5.5
UC-09	Escalate overdue complaint	Scheduler	5.6
UC-10	Close resolved complaint after the confirmation window	Scheduler	5.9
UC-11	View department work queue	Department Staff	5.7
UC-12	Accept and progress complaint	Department Staff	5.7
UC-13	Record resolution	Department Staff	5.7
UC-14	Reassign complaint	Department Head	5.7
UC-15	Monitor department performance	Department Head	5.10
UC-16	Track own complaints	Complainant	5.8
UC-17	Receive notification	Complainant, Staff, Head	5.8
UC-18	Confirm or reject resolution	Complainant	5.9
UC-19	Manage categories and departments	Administrator	5.10
UC-20	Manage routing rules and SLA thresholds	Administrator	5.10
UC-21	Manage users and role assignment	Administrator	5.1, 5.2
UC-22	View and export campus-wide reports	Administrator	5.10

4.3 Use case diagram

The diagram below shows the actors of Section 4.1 and the twenty-two use cases of Section 4.2. Complainant and Anonymous Complainant interact with the submission and tracking use cases; Department Staff and Department Head interact with the queue, assignment, resolution and monitoring use cases; and the Administrator interacts with the master-data and reporting use cases. Routing (UC-07) is performed by the system as an included step of complaint submission (UC-03 and UC-05) and is not initiated by the Scheduler. The Scheduler initiates only the two periodic runs: SLA escalation (UC-09) and closure of a resolved complaint after the confirmation window (UC-10). The Mail Service is a secondary actor that receives outbound notification requests.

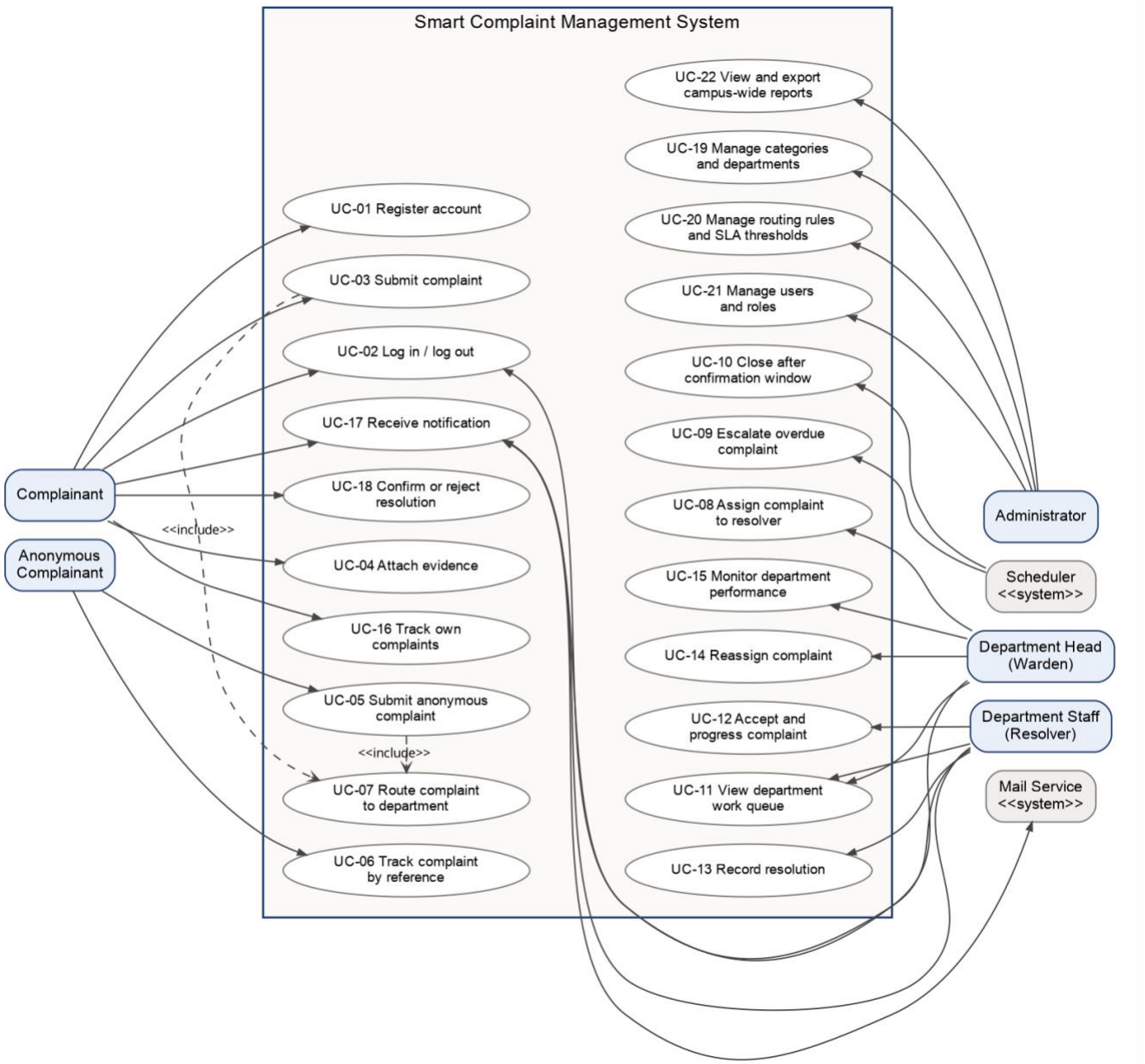


Figure 1 — SCMS use case diagram

4.4 Complaint state model

The complaint lifecycle is central to Section 5.7 and is specified normatively there. The permitted transitions are:

From state	Permitted next states	Permitted actor
`SUBMITTED`	`ASSIGNED`, `REJECTED`	System (routing), Administrator/Head (reject)
`ASSIGNED`	`IN_PROGRESS`, `REJECTED`	Assigned Staff, Head
`IN_PROGRESS`	`RESOLVED`	Assigned Staff
`RESOLVED`	`CLOSED`, `REOPENED`	Complainant (confirm/reject); system on auto-close timeout
`REOPENED`	`IN_PROGRESS`	Assigned Staff, Head
`CLOSED`	*(terminal)*	—
`REJECTED`	*(terminal)*	—

`ESCALATED` is **not** a state. It is an independent boolean flag with an escalation level, applicable to any complaint in `SUBMITTED`, `ASSIGNED`, `IN_PROGRESS` or `REOPENED`. This keeps the state machine free of combinatorial explosion and is a deliberate design-affecting decision recorded here.

Note on entity-relationship modelling. An ER diagram is applicable to this product but belongs to the Phase-3 design deliverable. Appendix B specifies the field-level data requirements that the eventual data model must satisfy.

5. System Features

Functional requirements are organised by system feature. Each requirement carries a unique tag of the form `REQ-<AREA>-<nn>`, and every tag appears in the traceability matrix in Appendix C. Feature ownership is recorded in the project synopsis.

5.1 User Registration and Authentication

Allows a user to create an account and authenticate. Every authenticated capability in the system depends on this feature, so it must be implemented first.

Priority: High. Benefit 9, Penalty 9, Cost 4, Risk 3.

5.1.1 Functional Requirements

ID	Requirement
REQ-AUTH-01	The system shall allow an unauthenticated visitor to register an account by supplying full name, email address, password, user type (student / faculty / staff) and, for hostel residents, hostel block and room. Field constraints are specified in Appendix B, Table B-1.
REQ-AUTH-02	The system shall reject a registration whose email address is already associated with an existing account without creating a duplicate account, and shall respond with the same generic acknowledgement used for an accepted registration, directing the user to the password-reset function if they already hold an account, so that no response discloses whether an account for that address exists (NFR-SEC-13).
REQ-AUTH-03	The system shall enforce a password policy of at least 8 characters including at least one letter and one digit, and shall reject a non-compliant password stating which conditions were not met.
REQ-AUTH-04	The system shall store passwords only as salted cryptographic hashes and shall never store, log, display or transmit a password in recoverable form.
REQ-AUTH-05	The system shall authenticate a user on presentation of a registered email and matching password, establish a session, and direct the user to the landing view of their assigned role. On failure it shall present a message that does not disclose which credential was incorrect.
REQ-AUTH-06	The system shall assign every newly registered account the Complainant role by default; no user shall be able to select or alter their own role at registration or thereafter.
REQ-AUTH-07	The system shall temporarily block authentication attempts for an account after 5 consecutive failures within 15 minutes, for a configurable lockout period defaulting to 15 minutes.
REQ-AUTH-08	The system shall terminate a session on explicit logout and after a configurable period of inactivity defaulting to 30 minutes, and shall require re-authentication for any subsequent action.

5.2 Role-Based Access Control

Restricts every action and every data view to the roles permitted to perform or see it. Authorisation is a graded security requirement for this project and a cross-cutting concern for all other features.

Priority: High. Benefit 9, Penalty 9, Cost 5, Risk 6.

5.2.1 Functional Requirements

ID	Requirement
REQ-RBAC-01	The system shall support exactly the roles Complainant, Department Staff, Department Head and Administrator, and shall associate every authenticated user with exactly one role.
REQ-RBAC-02	The system shall authorise every request against the requesting user's current role and, for department-scoped roles, their current department membership, and shall refuse any request that is not permitted, irrespective of how the request was constructed.
REQ-RBAC-03	The system shall permit a Complainant to read only complaints they themselves raised, and shall refuse read access to any other complaint.
REQ-RBAC-04	The system shall permit Department Staff to read and act only on complaints assigned to their own department, and shall refuse access to complaints of other departments.
REQ-RBAC-05	The system shall restrict management of users, roles, categories, departments, routing rules and SLA thresholds to the Administrator role only.
REQ-RBAC-06	The system shall record every refused authorisation attempt with the requesting account, the attempted action, the target resource and a timestamp.

5.3 Complaint Registration and Evidence Submission

Allows an authenticated complainant to lodge a complaint with structured classification and optional evidence, and returns a unique identifier as acknowledgement. This is the system's primary intake path.

Priority: High. Benefit 9, Penalty 9, Cost 5, Risk 4.

5.3.1 Functional Requirements

ID	Requirement
REQ-CMP-01	The system shall allow an authenticated Complainant to submit a complaint specifying category, sub-category where defined, location (block, floor, room), title, description and a priority indication. Field constraints are specified in Appendix B, Table B-2.
REQ-CMP-02	The system shall populate category and location selection lists only from master data currently marked active, and shall reject a submission referencing an inactive or unknown category or location.
REQ-CMP-03	The system shall require category, location, title and description, shall enforce a minimum description length of 20 characters, and shall refuse any submission not meeting these conditions while preserving the user's other entered values.
REQ-CMP-04	The system shall generate for every accepted complaint a unique, immutable, human-quotable identifier of the form `SCMS-YYYY-NNNNNN`, and shall display it on the acknowledgement.
REQ-CMP-05	The system shall allow a Complainant to attach up to 3 files to a complaint, restricted to types JPEG, PNG and PDF, each not exceeding 5 MB, and shall refuse the submission with a stated reason if these limits are exceeded.
REQ-CMP-06	The system shall validate an uploaded file's actual content type rather than relying on its filename extension or the client-declared type, and shall store it under a system-generated name that does not incorporate user-supplied text.
REQ-CMP-07	The system shall treat complaint creation, attachment storage and initial routing as a single atomic operation: if any part fails, no complaint, identifier or attachment shall persist.
REQ-CMP-08	The system shall record on every complaint the submitting account, the submission timestamp and the initial state `SUBMITTED`.
REQ-CMP-09	The system shall prevent a single submission action from creating more than one complaint, so that a retry after a timeout does not produce a duplicate.

5.4 Anonymous Complaint Submission and Tracking

Allows a complaint to be raised without login, and tracked afterwards using a system-issued reference. This exists to remove the attribution fear that suppresses reporting of sensitive issues.

Priority: Medium. Benefit 7, Penalty 5, Cost 4, Risk 5.

5.4.1 Functional Requirements

ID	Requirement
REQ-ANON-01	The system shall allow an unauthenticated visitor to submit a complaint, applying the same validation, identifier generation, attachment, atomicity and routing rules as REQ-CMP-01 through REQ-CMP-09, except for those requirements concerning the authenticated submitter account; REQ-ANON-03 is the authoritative requirement on what may be recorded for an anonymous complaint.
REQ-ANON-02	The system shall issue for every anonymous complaint a tracking reference that is unguessable, distinct from the complaint identifier, and displayed exactly once at submission with a notice that it cannot be recovered.
REQ-ANON-03	The system shall record no identifying data against an anonymous complaint — no account link, no name, no email address, and no client IP address — and no application function, view, export or report shall permit attribution of an anonymous complaint to an individual. Rate-limiting and security logging required by NFR-SEC-* shall operate on request metadata only and shall not store any value that links a request to an anonymous complaint record.
REQ-ANON-04	The system shall, on presentation of a valid tracking reference, disclose only that complaint's identifier, category, location, current status, escalation flag and status history, and nothing else.
REQ-ANON-05	The system shall throttle tracking lookups to at most 10 attempts per source per 10 minutes, and shall return an identical "no matching complaint" response for both unknown and malformed references. For throttling purposes, source shall mean the request-level rate-limit key used to group tracking attempts; this source value shall be used only for rate limiting and security controls and shall not be persisted or linked to the anonymous complaint record.
REQ-ANON-06	The system shall present an anonymous complaint to Staff, Head and Administrator users with the complainant shown as anonymous, and shall not offer any action that would attribute it to an individual.

5.5 Rule-Based Routing and Assignment

Determines the responsible department for each complaint from administrator-configured rules and places the complaint in that department's queue with no manual forwarding. This is the first half of the system's "smart" behaviour.

Priority: High. Benefit 9, Penalty 8, Cost 6, Risk 7.

5.5.1 Functional Requirements

ID	Requirement

ID	Requirement
REQ-ROUTE-01	The system shall determine the responsible department for every accepted complaint by evaluating active routing rules against the complaint's category and location, without any manual forwarding action.
REQ-ROUTE-02	The system shall resolve multiple matching rules by preferring the most specific match, in the order category-and-location, then category-only, then location-only; where specificity is equal it shall apply the rule with the highest configured precedence value.
REQ-ROUTE-03	The system shall assign any complaint matching no active rule to the configured default department and shall flag it as unrouted for administrator attention; no accepted complaint shall ever remain unassigned.
REQ-ROUTE-04	The system shall record for every routing decision the complaint, the resulting department, the identifier of the rule applied or an indication that the default was used, and the decision timestamp.
REQ-ROUTE-05	The system shall place every routed complaint in the receiving department's work queue and set its state to ASSIGNED, indicating that the complaint has been assigned to the department.
REQ-ROUTE-06	The system shall evaluate routing rules only against rules and departments currently marked active, and shall apply rule changes only to complaints submitted after the change.
REQ-ROUTE-07	The system shall order each department's work queue by escalation flag first, then by priority, then by age of submission, oldest first.
REQ-ROUTE-08	The system shall allow a Department Head to assign a department-assigned complaint in state ASSIGNED to a Department Staff member of the same department, shall record the assigning actor, the assignee and the timestamp in the complaint history per REQ-LIFE-04, and shall refuse assignment to a member of any other department. This individual resolver assignment shall not change the complaint state.

5.6 Priority, SLA Targets and Automatic Escalation

Attaches a resolution time target to every complaint and automatically escalates any complaint that breaches it. This is the second half of the "smart" behaviour and carries the highest technical risk in the project.

Priority: High. Benefit 9, Penalty 8, Cost 7, Risk 8.

5.6.1 Functional Requirements

ID	Requirement
REQ-SLA-01	The system shall assign every accepted complaint a priority from the set Low, Medium, High, Critical, taking the complainant's indication as input and overriding it where a configured category rule specifies a minimum priority.
REQ-SLA-02	The system shall compute and store for every accepted complaint a resolution due timestamp derived from its priority and category according to the SLA configuration held in master data.
REQ-SLA-03	The system shall provide an escalation evaluation run, invocable by an external scheduler, that identifies every complaint in state `SUBMITTED`, `ASSIGNED`, `IN_PROGRESS` or `REOPENED` whose resolution due timestamp has passed.
REQ-SLA-04	The system shall, for each such complaint, set its escalation flag, increment its escalation level up to the configured maximum, record the escalation with timestamp and reason in the status history, and notify the responsible Department Head. If the complaint is already at the configured maximum escalation level, the system shall retain that level and shall not increment it further. Repeated scheduler runs shall not create duplicate escalation events within the configured escalation interval.
REQ-SLA-05	The system shall not increase a complaint's escalation level more than once within one configured escalation interval, so that repeated or overlapping runs cannot produce duplicate escalations.
REQ-SLA-06	The system shall exclude complaints in state `RESOLVED`, `CLOSED` or `REJECTED` from escalation.
REQ-SLA-07	The system shall apply a change to SLA configuration only to complaints submitted after the change, leaving existing due timestamps unaltered.
REQ-SLA-08	The system shall make the escalation run resumable: a run that terminates abnormally shall leave the complaints it already processed correctly escalated and shall allow the remainder to be processed by the next run without double escalation.

5.7 Complaint Lifecycle Management and Audit Trail

Governs the states a complaint may occupy, who may move it between them, and records every transition immutably. This feature makes accountability possible and is the principal source of negative test cases.

Priority: High. Benefit 9, Penalty 9, Cost 6, Risk 6.

5.7.1 Functional Requirements

ID	Requirement
REQ-LIFE-01	The system shall maintain every complaint in exactly one of the states `SUBMITTED`, `ASSIGNED`, `IN_PROGRESS`, `RESOLVED`, `REOPENED`, `CLOSED`, `REJECTED`.
REQ-LIFE-02	The system shall permit only the state transitions defined in Section 4.4 and shall refuse any other transition, leaving the complaint's state, data and history unchanged and reporting that the transition is not permitted.
REQ-LIFE-03	The system shall permit a transition only to an actor authorised for it under Section 4.4 and Section 5.2, and shall refuse it otherwise.

ID	Requirement
REQ-LIFE-04	The system shall append to a complaint's status history, for every transition, escalation, reassignment and resolution entry, the acting account or the system, the previous state, the new state, the timestamp and any remarks supplied.
REQ-LIFE-05	The system shall make status history append-only, providing no interface by which any user, including an Administrator, can edit or delete a recorded history entry.
REQ-LIFE-06	The system shall present each Department Staff user a work queue of complaints assigned to their own department, ordered per REQ-ROUTE-07, and shall allow them to act only on complaints in that queue.
REQ-LIFE-07	The system shall require resolution remarks of at least 20 characters when a complaint is moved to RESOLVED, and shall allow resolution evidence to be attached subject to the limits in REQ-CMP-05.
REQ-LIFE-08	The system shall allow a Department Head to reassign a complaint that has been assigned to a Staff member under REQ-ROUTE-08 to another Department Staff member of the same department without changing its state, shall require a reassignment reason and store that reason with the reassignment actor and timestamp in the status history per REQ-LIFE-04, and shall refuse reassignment to a member of any other department.
REQ-LIFE-09	The system shall permit only one concurrent transition on a complaint, refusing a competing transition with an indication that the complaint has been modified, so that no complaint can reach an invalid or undefined state.
REQ-LIFE-10	The system shall prevent a Complainant from editing or transitioning a complaint once it has left state `SUBMITTED`, other than the confirm, reject and reopen actions specified in Section 5.9.

5.8 Complaint Tracking and Notifications

Gives the complainant continuous visibility of their complaint and informs all parties of significant state changes in-app and by email. This closes the feedback loop that the manual process leaves permanently open.

Priority: High. Benefit 8, Penalty 8, Cost 5, Risk 5.

5.8.1 Functional Requirements

ID	Requirement
REQ-NOTF-01	The system shall provide an authenticated Complainant a list of their own complaints showing identifier, category, location, current status, escalation flag, submission timestamp and resolution due timestamp, filterable by status and by category.
REQ-NOTF-02	The system shall provide, for any complaint the requesting user is authorised to read, a detail view containing all submitted data, attachments, current state, escalation level, assigned department and the complete status history in chronological order.
REQ-NOTF-03	The system shall create an in-app notification for the affected parties on successful complaint submission/acknowledgement, assignment to a department, commencement of work, resolution, escalation, reopening and closure.
REQ-NOTF-04	The system shall send an email notification, to any affected party having a recorded email address, on submission, assignment, resolution, escalation and closure, containing the complaint identifier and current status and complying with CI-6.
REQ-NOTF-05	The system shall treat email delivery as non-blocking: failure or unavailability of the mail interface shall not prevent or reverse the underlying state transition, and shall not prevent creation of the corresponding in-app notification.
REQ-NOTF-06	The system shall retry a failed email delivery up to 3 times with increasing delay, and shall record a permanent failure without altering complaint state.
REQ-NOTF-07	The system shall present a user's notifications newest first, distinguish unread from read, display an unread count in the header, and mark a notification read when it is viewed.

5.9 Resolution Feedback and Reopen

Requires the complainant to accept or reject a claimed resolution, and returns rejected complaints to the department queue. This prevents closure of problems that were never actually fixed.

Priority: High. Benefit 8, Penalty 7, Cost 4, Risk 4.

5.9.1 Functional Requirements

ID	Requirement
REQ-FDBK-01	The system shall allow only the originating Complainant, or the holder of the tracking reference for an anonymous complaint, to confirm or reject a resolution, and shall refuse the action to any other party.
REQ-FDBK-02	The system shall, on confirmation, move the complaint to `CLOSED`, record the confirming actor and timestamp, accept an optional satisfaction rating on a 1–5 scale, and notify the resolving department.
REQ-FDBK-03	The system shall, on rejection, require a reason of at least 20 characters, move the complaint to `REOPENED`, return it to the same department retaining its full history, increment the complaint's reopen count, and notify the department and its head.
REQ-FDBK-04	The system shall limit a complaint to a configurable maximum of 2 reopens; on a further rejection it shall refuse the reopen, raise the complaint to the Department Head for manual handling, and inform the complainant.
REQ-FDBK-05	The system shall automatically close a complaint that has remained in `RESOLVED` without complainant response for a configurable period defaulting to 7 days, and shall record in the status history that closure was automatic and unconfirmed.
REQ-FDBK-06	The system shall retain and display the reopen count and every rejection reason on the complaint detail view and in the reports of Section 5.10.

5.10 Master Data Administration and Reporting

Lets administrators configure the data that drives the system's behaviour, and turns accumulated complaints into information the administration can act on.

Priority: Medium–High. Benefit 8, Penalty 6, Cost 5, Risk 4.

5.10.1 Functional Requirements

ID	Requirement
REQ-ADMN-01	The system shall allow an Administrator to create, view, modify and deactivate complaint categories, sub-categories, locations and departments, and shall make active entries immediately available for new complaints.
REQ-ADMN-02	The system shall refuse deletion of any master-data entry referenced by an existing complaint, offering deactivation instead, and shall preserve the reference on complaints already using it.
REQ-ADMN-03	The system shall allow an Administrator to assign and remove Department Staff and Department Head membership of a department, and shall reflect the change in the affected user's queue and permissions on their next request.
REQ-ADMN-04	The system shall allow an Administrator to create, modify, reorder by precedence and deactivate routing rules mapping category and location to a department, and shall refuse a rule that duplicates an existing active rule.
REQ-ADMN-05	The system shall allow an Administrator to configure, per priority and optionally per category, the SLA resolution duration, the escalation interval and the maximum escalation level, accepting only positive durations and stating the permitted range on refusal.
REQ-ADMN-06	The system shall allow an Administrator to change any user's role, subject to at least one Administrator account always remaining.
REQ-ADMN-07	The system shall provide an Administrator a report, for a selectable date range, of complaint volume by category, by location and by department, average and median resolution time, and counts of open, overdue, escalated and reopened complaints.
REQ-ADMN-08	The system shall provide a Department Head the same measures as REQ-ADMN-07 restricted to their own department, and shall refuse them access to other departments' figures.
REQ-ADMN-09	The system shall record every master-data change with the acting Administrator, the entity changed, the previous and new values and a timestamp.
REQ-ADMN-10	The system shall allow an Administrator to export the report generated under REQ-ADMN-07 for the selected date range as a CSV file, containing the same figures as the on-screen report, and shall allow a Department Head to export the report of REQ-ADMN-08 restricted to their own department.

6. Other Nonfunctional Requirements

6.1 Performance Requirements

ID	Requirement
NFR-PERF-01	95% of complaint submission, complaint list, complaint detail and tracking requests shall complete within **2 seconds**, measured server-side at a sustained load of 50 concurrent users on the target deployment.
NFR-PERF-02	95% of staff work-queue and complaint-action requests shall complete within **1.5 seconds** under the same load.
NFR-PERF-03	The escalation evaluation run shall process 5,000 open complaints within **60 seconds**.
NFR-PERF-04	Report generation over a 12-month range shall complete within **5 seconds**.
NFR-PERF-05	An attachment of up to 5 MB shall be accepted, and the acknowledgement returned, within **30 seconds** on a 2 Mbps uplink.
NFR-PERF-06	No user-facing list view shall load more than 25 records per request.

6.2 Safety Requirements

SCMS controls no physical process and can cause no direct physical harm. The safety concerns are those of a system whose failure allows a real hazard to go unattended.

ID	Requirement
NFR-SAFE-01	The system shall never silently discard an accepted complaint: once acknowledged, a complaint shall be recoverable and visible in some queue until it reaches a terminal state. A complaint matching no routing rule shall go to the default queue flagged for attention, never to nowhere (REQ-ROUTE-03).
NFR-SAFE-02	The system shall not present itself as an emergency reporting channel. Every complaint submission screen shall display a notice directing users to the appropriate emergency contact for fire, electrical danger, medical emergency or structural hazard.
NFR-SAFE-03	Failure of the escalation run shall be detectable: the system shall record the start, completion and outcome of every run so that a silent failure to escalate cannot go unnoticed.

ID	Requirement
NFR-SAFE-04	The system shall preserve all complaint and status-history data through application restart, deployment and container replacement; no accepted complaint shall be lost by an operational action.
NFR-SAFE-05	The system shall not permit deletion of a complaint by any role. Complaints are terminated only by reaching `CLOSED` or `REJECTED`.

6.3 Security Requirements

ID	Requirement
NFR-SEC-01	The system shall authenticate every user before granting access to any function other than registration, login and anonymous complaint submission and tracking.
NFR-SEC-02	The system shall store passwords only as salted cryptographic hashes using a currently accepted algorithm (REQ-AUTH-04), and shall never write a password or session token to any log.
NFR-SEC-03	The system shall enforce authorisation server-side on every request, independently of the user interface presented. Absence of a link or button shall never be the only control preventing an action.
NFR-SEC-04	The system shall prevent horizontal privilege escalation: no user shall read or act on another user's complaint by manipulating an identifier in a URL or form field (REQ-RBAC-03, REQ-RBAC-04).
NFR-SEC-05	The system shall validate and sanitise all user-supplied input server-side, and shall escape all user-supplied content on output so that stored or reflected script cannot execute.
NFR-SEC-06	The system shall construct all database access through the ORM's parameterised mechanisms, so that no user input can alter query structure.
NFR-SEC-07	The system shall require a valid anti-CSRF token on every state-changing request (CI-3).
NFR-SEC-08	The system shall transmit all traffic over TLS and set session cookies `Secure`, `HttpOnly` and `SameSite` (CI-1, CI-4).
NFR-SEC-09	The system shall restrict uploaded files by verified content type and size (REQ-CMP-05, REQ-CMP-06), shall store them outside the web root or serve them only through an authorisation check, and shall never execute an uploaded file.
NFR-SEC-10	The system shall serve an attachment only to a user authorised to read the parent complaint; a direct file reference shall not bypass that check.
NFR-SEC-11	The system shall preserve anonymity for anonymous complaints as specified in REQ-ANON-03: no application data, view, export or report shall permit attribution of an anonymous complaint to an individual.
NFR-SEC-12	The system shall throttle authentication attempts (REQ-AUTH-07) and anonymous tracking lookups (REQ-ANON-05) to resist brute-force enumeration.
NFR-SEC-13	The system shall return responses that do not disclose whether an account, complaint identifier or tracking reference exists, beyond what the requesting user is authorised to know.
NFR-SEC-14	The system shall keep all secrets — database credentials, mail credentials, application signing key — in environment configuration, never in source control (SI-R1).
NFR-SEC-15	The system shall run with debug output disabled in the deployed environment, and shall present generic error pages carrying no internal detail (UI-6).
NFR-SEC-16	The system shall record security-relevant events — authentication success and failure, lockout, authorisation refusal (REQ-RBAC-06), role change, master-data change — with actor, action, target and timestamp.

6.4 Software Quality Attributes

Stated in preference order: **correctness and security first, then reliability, then usability, then maintainability, then performance beyond the stated thresholds.** Where a trade-off arises, correctness of the complaint record takes precedence over responsiveness.

ID	Requirement
NFR-QUAL-01	100% of complaints matching an active routing rule shall be assigned to the department that rule specifies; 100% of complaints matching no rule shall reach the default queue.
NFR-QUAL-02	A retry after a client timeout shall not create a duplicate complaint (REQ-CMP-09), and no complaint shall ever occupy an invalid or undefined state (REQ-LIFE-02, REQ-LIFE-09).
NFR-QUAL-03	Every complaint breaching its SLA shall be flagged and escalated within one escalation interval of the breach (REQ-SLA-03, REQ-SLA-04).
NFR-QUAL-04	A first-time complainant unfamiliar with the system shall complete registration and a full complaint submission unaided, traversing no more than four primary screens (UI-4).
NFR-QUAL-05	Every refusal shall state what was wrong and what the user must do, in language free of technical jargon.
NFR-QUAL-06	Every state transition, escalation, reassignment and master-data change shall be recorded with actor and timestamp, and shall not be editable or deletable through the application (REQ-LIFE-04, REQ-LIFE-05).
NFR-QUAL-07	Adding a complaint category, sub-category, location, department, routing rule or SLA threshold shall require configuration only and no code change (C-12).

ID	Requirement
NFR-QUAL-08	Each of the four functional areas F1–F4 shall be implemented as an identifiable, separately unit-testable module with business logic outside presentation templates (C-8, C-11).
NFR-QUAL-09	Every functional requirement in Section 5 shall be verifiable by at least one automated test, and every state transition rule shall have both a positive and a negative test.
NFR-QUAL-10	Unit tests shall achieve at least **70% line coverage** and **60% branch coverage** across the four functional modules, reported by the coverage tool (C-7).
NFR-QUAL-11	The application shall run unchanged, driven only by configuration, in a local development environment on SQLite and in the deployed container on PostgreSQL.
NFR-QUAL-12	The system shall be designed for restart without data loss (NFR-SAFE-04). No availability percentage is committed for Release 1.0, this being a student deployment.

6.5 Business Rules

ID	Business rule
BR-01	Every complaint belongs to exactly one department at any time and is never unassigned once accepted.
BR-02	Only the assigned department's staff, or its head, may progress a complaint.
BR-03	Only the originating complainant may accept or reject a resolution; a resolver may never close their own work.
BR-04	A complaint is closed only by complainant confirmation or by expiry of the confirmation window.
BR-05	A complaint may be reopened at most twice; beyond that it becomes the Department Head's responsibility.
BR-06	Complaint history is a permanent record. Nobody, including an Administrator, may alter or remove it.
BR-07	A user may never assign or change their own role.
BR-08	At least one Administrator account must exist at all times.
BR-09	An anonymous complaint must never become attributable to an individual, by any function or report.
BR-10	Master data referenced by a complaint is deactivated, never deleted.
BR-11	Routing and SLA configuration changes apply prospectively only; a complaint is judged by the rules in force when it was submitted.
BR-12	Priority indicated by a complainant may be raised by a configured category rule but the system's derived priority governs the SLA.
BR-13	Escalation raises visibility and notifies authority; it never reassigns the complaint automatically.
BR-14	SCMS records the workflow only; the physical repair occurs outside the system.

Domain requirements

ID	Domain requirement
DOM-01	Complaint categories reflect the campus facility domain: Electrical, Plumbing and Water Supply, Network and IT, Housekeeping, Furniture and Fittings, Mess and Canteen, Civil and Structural, Other. Exact set subject to stakeholder confirmation (D-A4).
DOM-02	Location is expressed in campus terms as block, floor and room, not as coordinates.
DOM-03	Departmental structure mirrors the campus maintenance organisation, one department per maintenance trade, and must be reconfigurable without code change.
DOM-04	SLA durations must be expressible in hours and days, and reflect realistic campus maintenance turnaround rather than commercial help-desk norms. Values subject to stakeholder confirmation (D-A4).
DOM-05	The academic calendar affects complaint volume and staff availability. Release 1.0 does not model working hours or holidays; SLA durations are elapsed-time based. This is a known simplification.

Appendix A: Glossary

Term	Definition
ABI	Application Binary Interface.
Anonymous Complainant	A person who submits a complaint without authenticating; identified to the system only by a tracking reference.
API	Application Programming Interface.
Assignee	The Department Staff member currently responsible for working a complaint.
Attachment	A file of evidence associated with a complaint at submission or at resolution.
Audit trail	The append-only status history of a complaint; see *Status history*.
Complainant	An authenticated user who raises a complaint.
Complaint	A recorded report of a facility or hostel problem, uniquely identified and progressed through the lifecycle of Section 4.4.
Complaint identifier	The unique, immutable, human-quotable reference of the form `SCMS-YYYY-NNNNNN` issued at acceptance.
CSRF	Cross-Site Request Forgery.

Term	Definition
Default department	The department to which a complaint matching no routing rule is assigned, flagged for administrator attention.
Escalation	Raising a complaint's visibility and notifying higher authority because its SLA target has been breached. Represented as a flag with a level, not as a state.
Escalation interval	The minimum period between successive escalation-level increases on one complaint.
F1–F4	The four functional ownership areas defined in the synopsis: F1 Foundation and Access; F2 Complaint Intake and Evidence; F3 Routing, Assignment and Escalation; F4 Resolution, Notifications and Tracking. Administrative reporting is delivered within F1 and F3 as recorded in the feature ownership of the synopsis.
Master data	Administrator-configured reference data: categories, sub-categories, locations, departments, routing rules and SLA configuration.
ORM	Object-Relational Mapper; the sole permitted means of database access (SI-1).
Priority	One of Low, Medium, High, Critical; determines the SLA duration applied.
RBAC	Role-Based Access Control.
Reopen	Return of a `RESOLVED` complaint to `REOPENED` because the complainant rejected the resolution.
Resolution due timestamp	The point in time by which a complaint must reach `RESOLVED`, derived from priority and category at acceptance.
Resolver	Synonym for Department Staff.
Routing	Automatic determination of the responsible department from configured rules.
Routing rule	A configured mapping from category and/or location to a department, with a precedence value.
SCMS	Smart Complaint Management System.
SLA	Service Level Agreement; here, the configured resolution time target for a priority and category.
SRS	Software Requirements Specification.
Status history	The append-only record of every transition, escalation, reassignment and resolution on a complaint, each with actor, timestamp and remarks.
Tracking reference	The unguessable token issued once for an anonymous complaint, permitting status lookup without authentication.
Unrouted flag	Marker on a complaint that reached the default department because no rule matched.
Work queue	The ordered list of complaints assigned to a department, sorted by escalation, then priority, then age.
Source	The request-level rate-limit key used to group anonymous tracking attempts; it is not persisted or linked to an anonymous complaint record.

Appendix B: Field Layouts

Field layouts and report field requirements. These define the data-level requirements the eventual design must satisfy; they are not a database schema. The raw password is an input-only value: it is validated against the password rules and then discarded, and only the salted cryptographic hash is persisted, in the Password Hash field of Table B-1 (REQ-AUTH-04). A raw password is never stored, logged, displayed or transmitted in recoverable form.

Table B-1 — User registration

Field	Data type and length	Is Mandatory
Full Name	String (100)	Y
Email Address	String (120)	Y
Password Hash	String (128)	Y
User Type	Alphanumeric (20)	Y
Contact Number	Numeric (15)	N
Hostel Block	Alphanumeric (20)	N
Room Number	Alphanumeric (10)	N
Role	Alphanumeric (20)	Y
Department	Alphanumeric (50)	N
Account Status	Alphanumeric (15)	Y
Registered On	Date (8)	Y

Table B-2 — Complaint registration

Field	Data type and length	Is Mandatory
Complaint Identifier	Alphanumeric (18)	Y
Complainant Reference	Reference (—)	N
Tracking Reference	Alphanumeric (32)	N
Category	Alphanumeric (50)	Y
Sub-Category	Alphanumeric (50)	N
Block	Alphanumeric (20)	Y
Floor	Alphanumeric (10)	N
Room / Area	Alphanumeric (30)	Y
Title	String (120)	Y
Description	String (2000)	Y
Indicated Priority	Alphanumeric (10)	Y
Derived Priority	Alphanumeric (10)	Y
Assigned Department	Alphanumeric (50)	Y
Applied Routing Rule	Reference (—)	Y
Unrouted Flag	Boolean (1)	Y
Assignee	Reference (—)	N
Current State	Alphanumeric (15)	Y
Escalation Flag	Boolean (1)	Y
Escalation Level	Numeric (2)	Y
Submitted On	DateTime (14)	Y
Resolution Due On	DateTime (14)	Y
Resolved On	DateTime (14)	N
Closed On	DateTime (14)	N
Resolution Remarks	String (2000)	N
Reopen Count	Numeric (1)	Y
Satisfaction Rating	Numeric (1)	N

Table B-3 — Complaint attachment

Field	Data type and length	Is Mandatory
Attachment Identifier	Alphanumeric (36)	Y
Complaint Reference	Reference (—)	Y
Stored File Name	String (100)	Y
Original File Name	String (255)	Y
Verified Content Type	Alphanumeric (30)	Y
File Size	Numeric (10)	Y
Attachment Stage	Alphanumeric (15)	Y
Uploaded By	Reference (—)	N
Uploaded On	DateTime (14)	Y

Table B-4 — Routing rule

Field	Data type and length	Is Mandatory
Rule Identifier	Alphanumeric (10)	Y
Category	Alphanumeric (50)	N
Block	Alphanumeric (20)	N
Target Department	Alphanumeric (50)	Y
Precedence	Numeric (3)	Y
Is Active	Boolean (1)	Y
Created By	Reference (—)	Y
Created On	DateTime (14)	Y

Table B-5 — SLA configuration

Field	Data type and length	Is Mandatory
Configuration Identifier	Alphanumeric (10)	Y
Priority	Alphanumeric (10)	Y
Category	Alphanumeric (50)	N
Resolution Duration (hours)	Numeric (4)	Y
Escalation Interval (hours)	Numeric (4)	Y
Maximum Escalation Level	Numeric (2)	Y
Minimum Priority Override	Alphanumeric (10)	N
Is Active	Boolean (1)	Y

Table B-6 — Status history entry

Field	Data type and length	Is Mandatory
History Identifier	Alphanumeric (36)	Y
Complaint Reference	Reference (—)	Y
Entry Type	Alphanumeric (20)	Y
Previous State	Alphanumeric (15)	N
New State	Alphanumeric (15)	Y
Acting Account	Reference (—)	N
Actor Kind	Alphanumeric (10)	Y
Remarks	String (2000)	N
Recorded On	DateTime (14)	Y

Appendix C: Requirement Traceability Matrix

This matrix contains every identified requirement in this document: the functional requirements of Section 5 (REQ-*), the non-functional requirements of Section 6 covering performance, safety, security and software quality (NFR-*), and the business rules and domain requirements of Section 6.5 (BR- *and DOM-*), which are treated as formal traceable requirements. It therefore holds 117 rows, one for every requirement identifier used in the document. Architecture, design, code and test references are recorded as **TBD** at SRS baseline and are populated as the corresponding phases complete. Every row must be closed before the final report.

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
1	REQ-AUTH-01	Allow an unauthenticated visitor to register an account by…	TBD	TBD	TBD	TBD	TBD
2	REQ-AUTH-02	Reject a registration whose email address is already associated…	TBD	TBD	TBD	TBD	TBD
3	REQ-AUTH-03	Enforce a password policy of at least 8 characters…	TBD	TBD	TBD	TBD	TBD
4	REQ-AUTH-04	Store passwords only as salted cryptographic hashes and shall…	TBD	TBD	TBD	TBD	TBD
5	REQ-AUTH-05	Authenticate a user on presentation of a registered email…	TBD	TBD	TBD	TBD	TBD
6	REQ-AUTH-06	Assign every newly registered account the Complainant role by…	TBD	TBD	TBD	TBD	TBD
7	REQ-AUTH-07	Temporarily block authentication attempts for an account after 5…	TBD	TBD	TBD	TBD	TBD
8	REQ-AUTH-08	Terminate a session on explicit logout and after a…	TBD	TBD	TBD	TBD	TBD
9	REQ-RBAC-01	Support exactly the roles Complainant, Department Staff, Department Head…	TBD	TBD	TBD	TBD	TBD
10	REQ-RBAC-02	Authorise every request against the requesting user's current role…	TBD	TBD	TBD	TBD	TBD
11	REQ-RBAC-03	Permit a Complainant to read only complaints they themselves…	TBD	TBD	TBD	TBD	TBD
12	REQ-RBAC-04	Permit Department Staff to read and act only on…	TBD	TBD	TBD	TBD	TBD
13	REQ-RBAC-05	Restrict management of users, roles, categories, departments, routing rules…	TBD	TBD	TBD	TBD	TBD

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
14	REQ-RBAC-06	Record every refused authorisation attempt with the requesting account,…	TBD	TBD	TBD	TBD	TBD
15	REQ-CMP-01	Allow an authenticated Complainant to submit a complaint specifying…	TBD	TBD	TBD	TBD	TBD
16	REQ-CMP-02	Populate category and location selection lists only from master…	TBD	TBD	TBD	TBD	TBD
17	REQ-CMP-03	Require category, location, title and description, shall enforce a…	TBD	TBD	TBD	TBD	TBD
18	REQ-CMP-04	Generate for every accepted complaint a unique, immutable, human-quotable…	TBD	TBD	TBD	TBD	TBD
19	REQ-CMP-05	Allow a Complainant to attach up to 3 files…	TBD	TBD	TBD	TBD	TBD
20	REQ-CMP-06	Validate an uploaded file's actual content type rather than…	TBD	TBD	TBD	TBD	TBD
21	REQ-CMP-07	Treat complaint creation, attachment storage and initial routing as…	TBD	TBD	TBD	TBD	TBD
22	REQ-CMP-08	Record on every complaint the submitting account, the submission…	TBD	TBD	TBD	TBD	TBD
23	REQ-CMP-09	Prevent a single submission action from creating more than…	TBD	TBD	TBD	TBD	TBD
24	REQ-ANON-01	Allow an unauthenticated visitor to submit a complaint, applying…	TBD	TBD	TBD	TBD	TBD
25	REQ-ANON-02	Issue for every anonymous complaint a tracking reference that…	TBD	TBD	TBD	TBD	TBD
26	REQ-ANON-03	Record no identifying data against an anonymous complaint —…	TBD	TBD	TBD	TBD	TBD
27	REQ-ANON-04	The system shall, on presentation of a valid tracking…	TBD	TBD	TBD	TBD	TBD
28	REQ-ANON-05	Throttle tracking lookups to at most 10 attempts per…	TBD	TBD	TBD	TBD	TBD
29	REQ-ANON-06	Present an anonymous complaint to Staff, Head and Administrator…	TBD	TBD	TBD	TBD	TBD
30	REQ-ROUTE-01	Determine the responsible department for every accepted complaint by…	TBD	TBD	TBD	TBD	TBD
31	REQ-ROUTE-02	Resolve multiple matching rules by preferring the most specific…	TBD	TBD	TBD	TBD	TBD
32	REQ-ROUTE-03	Assign any complaint matching no active rule to the…	TBD	TBD	TBD	TBD	TBD
33	REQ-ROUTE-04	Record for every routing decision the complaint, the resulting…	TBD	TBD	TBD	TBD	TBD
34	REQ-ROUTE-05	Place routed complaint in department queue and set ASSIGNED to indicate departmental assignment	TBD	TBD	TBD	TBD	TBD
35	REQ-ROUTE-06	Evaluate routing rules only against rules and departments currently…	TBD	TBD	TBD	TBD	TBD
36	REQ-ROUTE-07	Order each department's work queue by escalation flag first,…	TBD	TBD	TBD	TBD	TBD
37	REQ-ROUTE-08	Department Head assigns department-assigned complaint to a Staff resolver within same department	TBD	TBD	TBD	TBD	TBD
38	REQ-SLA-01	Assign every accepted complaint a priority from the set…	TBD	TBD	TBD	TBD	TBD
39	REQ-SLA-02	Compute and store for every accepted complaint a resolution…	TBD	TBD	TBD	TBD	TBD
40	REQ-SLA-03	Provide an escalation evaluation run, invocable by an external…	TBD	TBD	TBD	TBD	TBD
41	REQ-SLA-04	Set escalation flag, increase level up to maximum, record and notify Head; retain max without further increment	TBD	TBD	TBD	TBD	TBD
42	REQ-SLA-05	Not increase a complaint's escalation level more than once…	TBD	TBD	TBD	TBD	TBD
43	REQ-SLA-06	Exclude complaints in state RESOLVED, CLOSED or REJECTED from…	TBD	TBD	TBD	TBD	TBD
44	REQ-SLA-07	Apply a change to SLA configuration only to complaints…	TBD	TBD	TBD	TBD	TBD
45	REQ-SLA-08	Make the escalation run resumable: a run that terminates…	TBD	TBD	TBD	TBD	TBD
46	REQ-LIFE-01	Maintain every complaint in exactly one of the states…	TBD	TBD	TBD	TBD	TBD
47	REQ-LIFE-02	Permit only the state transitions defined in Section 4.4…	TBD	TBD	TBD	TBD	TBD
48	REQ-LIFE-03	Permit a transition only to an actor authorised for…	TBD	TBD	TBD	TBD	TBD
49	REQ-LIFE-04	Append to a complaint's status history, for every transition,…	TBD	TBD	TBD	TBD	TBD
50	REQ-LIFE-05	Make status history append-only, providing no interface by which…	TBD	TBD	TBD	TBD	TBD
51	REQ-LIFE-06	Present each Department Staff user a work queue of…	TBD	TBD	TBD	TBD	TBD
52	REQ-LIFE-07	Require resolution remarks and allow resolution evidence on RESOLVED	TBD	TBD	TBD	TBD	TBD
53	REQ-LIFE-08	Reassign assigned complaint to another same-department Staff member with reason and audit entry	TBD	TBD	TBD	TBD	TBD

Sl. No	Requirement ID	Brief Description of Requirement	Architecture Reference	Design Reference	Code File Reference	Test Case ID	System Test Case ID
54	REQ-LIFE-09	Permit only one concurrent transition on a complaint, refusing…	TBD	TBD	TBD	TBD	TBD
55	REQ-LIFE-10	Prevent a Complainant from editing or transitioning a complaint…	TBD	TBD	TBD	TBD	TBD
56	REQ-NOTF-01	Provide an authenticated Complainant a list of their own…	TBD	TBD	TBD	TBD	TBD
57	REQ-NOTF-02	Provide, for any complaint the requesting user is authorised…	TBD	TBD	TBD	TBD	TBD
58	REQ-NOTF-03	Create in-app notifications on successful submission/acknowledgement and significant lifecycle events	TBD	TBD	TBD	TBD	TBD
59	REQ-NOTF-04	Send an email notification, to any affected party having…	TBD	TBD	TBD	TBD	TBD
60	REQ-NOTF-05	Treat email delivery as non-blocking: failure or unavailability of…	TBD	TBD	TBD	TBD	TBD
61	REQ-NOTF-06	Retry a failed email delivery up to 3 times…	TBD	TBD	TBD	TBD	TBD
62	REQ-NOTF-07	Present a user's notifications newest first, distinguish unread from…	TBD	TBD	TBD	TBD	TBD
63	REQ-FDBK-01	Allow only the originating Complainant, or the holder of…	TBD	TBD	TBD	TBD	TBD
64	REQ-FDBK-02	The system shall, on confirmation, move the complaint to…	TBD	TBD	TBD	TBD	TBD
65	REQ-FDBK-03	The system shall, on rejection, require a reason of…	TBD	TBD	TBD	TBD	TBD
66	REQ-FDBK-04	Limit a complaint to a configurable maximum of 2…	TBD	TBD	TBD	TBD	TBD
67	REQ-FDBK-05	Automatically close a complaint that has remained in RESOLVED…	TBD	TBD	TBD	TBD	TBD
68	REQ-FDBK-06	Retain and display the reopen count and every rejection…	TBD	TBD	TBD	TBD	TBD
69	REQ-ADMN-01	Allow an Administrator to create, view, modify and deactivate…	TBD	TBD	TBD	TBD	TBD
70	REQ-ADMN-02	Refuse deletion of any master-data entry referenced by an…	TBD	TBD	TBD	TBD	TBD
71	REQ-ADMN-03	Allow an Administrator to assign and remove Department Staff…	TBD	TBD	TBD	TBD	TBD
72	REQ-ADMN-04	Allow an Administrator to create, modify, reorder by precedence…	TBD	TBD	TBD	TBD	TBD
73	REQ-ADMN-05	Allow an Administrator to configure, per priority and optionally…	TBD	TBD	TBD	TBD	TBD
74	REQ-ADMN-06	Allow an Administrator to change any user's role, subject…	TBD	TBD	TBD	TBD	TBD
75	REQ-ADMN-07	Provide an Administrator a report, for a selectable date…	TBD	TBD	TBD	TBD	TBD
76	REQ-ADMN-08	Provide a Department Head the same measures as REQ-ADMN-07…	TBD	TBD	TBD	TBD	TBD
77	REQ-ADMN-09	Record every master-data change with the acting Administrator, the…	TBD	TBD	TBD	TBD	TBD
78	REQ-ADMN-10	Allow an Administrator to export the report generated under…	TBD	TBD	TBD	TBD	TBD
79	NFR-PERF-01	95% of complaint submission, complaint list, complaint detail and…	TBD	TBD	TBD	TBD	TBD
80	NFR-PERF-02	95% of staff work-queue and complaint-action requests shall complete…	TBD	TBD	TBD	TBD	TBD
81	NFR-PERF-03	The escalation evaluation run shall process 5,000 open complaints…	TBD	TBD	TBD	TBD	TBD
82	NFR-PERF-04	Report generation over a 12-month range shall complete within…	TBD	TBD	TBD	TBD	TBD
83	NFR-PERF-05	An attachment of up to 5 MB shall be…	TBD	TBD	TBD	TBD	TBD
84	NFR-PERF-06	No user-facing list view shall load more than 25…	TBD	TBD	TBD	TBD	TBD
85	NFR-SAFE-01	Never silently discard an accepted complaint: once acknowledged, a…	TBD	TBD	TBD	TBD	TBD
86	NFR-SAFE-02	Not present itself as an emergency reporting channel. Every…	TBD	TBD	TBD	TBD	TBD
87	NFR-SAFE-03	Failure of the escalation run shall be detectable: the…	TBD	TBD	TBD	TBD	TBD
88	NFR-SAFE-04	Preserve all complaint and status-history data through application restart,…	TBD	TBD	TBD	TBD	TBD
89	NFR-SAFE-05	Not permit deletion of a complaint by any role.…	TBD	TBD	TBD	TBD	TBD
90	NFR-SEC-01	Authenticate every user before granting access to any function…	TBD	TBD	TBD	TBD	TBD
91	NFR-SEC-02	Store passwords only as salted cryptographic hashes using a…	TBD	TBD	TBD	TBD	TBD
92	NFR-SEC-03	Enforce authorisation server-side on every request, independently of the…	TBD	TBD	TBD	TBD	TBD
93	NFR-SEC-04	Prevent horizontal privilege escalation: no user shall read or…	TBD	TBD	TBD	TBD	TBD
94	NFR-SEC-05	Validate and sanitise all user-supplied input server-side, and shall…	TBD	TBD	TBD	TBD	TBD

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95	NFR-SEC-06	Construct all database access through the ORM's parameterised mechanisms,…	TBD	TBD	TBD	TBD	TBD
96	NFR-SEC-07	Require a valid anti-CSRF token on every state-changing request…	TBD	TBD	TBD	TBD	TBD
97	NFR-SEC-08	Transmit all traffic over TLS and set session cookies…	TBD	TBD	TBD	TBD	TBD
98	NFR-SEC-09	Restrict uploaded files by verified content type and size…	TBD	TBD	TBD	TBD	TBD
99	NFR-SEC-10	Serve an attachment only to a user authorised to…	TBD	TBD	TBD	TBD	TBD
100	NFR-SEC-11	Preserve anonymity for anonymous complaints as specified in REQ-ANON-03:…	TBD	TBD	TBD	TBD	TBD
101	NFR-SEC-12	Throttle authentication attempts (REQ-AUTH-07) and anonymous tracking lookups (REQ-ANON-05)…	TBD	TBD	TBD	TBD	TBD
102	NFR-SEC-13	Return responses that do not disclose whether an account,…	TBD	TBD	TBD	TBD	TBD
103	NFR-SEC-14	Keep all secrets — database credentials, mail credentials, application…	TBD	TBD	TBD	TBD	TBD
104	NFR-SEC-15	Run with debug output disabled in the deployed environment,…	TBD	TBD	TBD	TBD	TBD
105	NFR-SEC-16	Record security-relevant events — authentication success and failure, lockout,…	TBD	TBD	TBD	TBD	TBD
106	NFR-QUAL-01	100% of complaints matching an active routing rule shall…	TBD	TBD	TBD	TBD	TBD
107	NFR-QUAL-02	A retry after a client timeout shall not create…	TBD	TBD	TBD	TBD	TBD
108	NFR-QUAL-03	Every complaint breaching its SLA shall be flagged and…	TBD	TBD	TBD	TBD	TBD
109	NFR-QUAL-04	A first-time complainant unfamiliar with the system shall complete…	TBD	TBD	TBD	TBD	TBD
110	NFR-QUAL-05	Every refusal shall state what was wrong and what…	TBD	TBD	TBD	TBD	TBD
111	NFR-QUAL-06	Every state transition, escalation, reassignment and master-data change shall…	TBD	TBD	TBD	TBD	TBD
112	NFR-QUAL-07	Adding a complaint category, sub-category, location, department, routing rule…	TBD	TBD	TBD	TBD	TBD
113	NFR-QUAL-08	Each of the four functional areas F1–F4 shall be…	TBD	TBD	TBD	TBD	TBD
114	NFR-QUAL-09	Every functional requirement in Section 5 shall be verifiable…	TBD	TBD	TBD	TBD	TBD
115	NFR-QUAL-10	Unit tests shall achieve at least 70% line coverage…	TBD	TBD	TBD	TBD	TBD
116	NFR-QUAL-11	The application shall run unchanged, driven only by configuration,…	TBD	TBD	TBD	TBD	TBD
117	NFR-QUAL-12	Be designed for restart without data loss (NFR-SAFE-04). No…	TBD	TBD	TBD	TBD	TBD